

O & Soergel, lec 8: Basics on category $\mathcal{O}$, IV

- 1) Equivalence between infinitesimal blocks
- 2) Structure of infinitesimal blocks.

1) Equivalence between infinitesimal blocks

We continue to assume that A is a local Noetherian algebra, $X \in \mathfrak{g}^* \otimes \mathfrak{m}$.

Our goal in this section is to prove:

Thm: Let $\lambda_1, \lambda_2 \in \Lambda$ be s.t. $\lambda_i + \rho$ is dominant. Assume

$$W_0 := \text{Stab}_W(\lambda_1 + \rho) = \text{Stab}_W(\lambda_2 + \rho).$$

Then $\exists$ equivalence of abelian categories $\mathcal{O}_A^{W \cdot \lambda_1} \xrightarrow{\sim} \mathcal{O}_A^{W \cdot \lambda_2}$

$$w. \Delta_A(w \cdot \lambda_1) \mapsto \Delta_A(w \cdot \lambda_2) \quad \forall w \in W.$$

An important ingredient of the construction & subsequent developments is a construction of a

$\overline{1}$ projective generator $P \in \mathcal{O}_A^{W \cdot \lambda}$ (Sec 2.3 of Lec 7)

(♥) for suitable dominant λ'

$$P = \mathcal{H}_{\Delta_A}(\lambda'), \quad \mathcal{H} := \mathcal{T}_{\lambda' \rightarrow \lambda}^\circ \left(\bigoplus_{w \in W} \bigotimes_w \right).$$

Proof of Thm: First note that $\exists \lambda_3 \in \Lambda$ s.t.

$$\text{Stab}_W(\lambda_3 + \rho) = W_0 \text{ \& } \lambda_3 - \lambda_i \text{ is dominant } (\forall i=1,2).$$

($\lambda_3 = \sum a_i \omega_i$ w. $a_i \gg 0$ & sum over all i w. ω_i fixed by W_0).

We claim that $\mathcal{F} := \mathcal{T}_{\lambda_i \rightarrow \lambda_3} : \mathcal{O}_A^{w \cdot \lambda_i} \rightarrow \mathcal{O}_A^{w \cdot \lambda_3}$ is an equivalence.

The functor $\mathcal{G} := \mathcal{T}_{\lambda_3 \rightarrow \lambda_i}$ is biadjoint to $\mathcal{F} \rightsquigarrow$ adjunc-

tion unit $\eta : \text{id} \rightarrow \mathcal{F}\mathcal{G}$ & counit $\varepsilon : \mathcal{G}\mathcal{F} \rightarrow \text{id}$. We need to prove

(*) η, ε are functor isomorphisms.

By axioms of adjunction, the compositions

$$(*) (*) \quad \mathcal{G} \xrightarrow{1_{\mathcal{G}}} \mathcal{G}\mathcal{F}\mathcal{G} \xrightarrow{\varepsilon 1_{\mathcal{G}}} \mathcal{G} \text{ \& } \mathcal{F} \xrightarrow{1_{\mathcal{F}}} \mathcal{F}\mathcal{G}\mathcal{F} \xrightarrow{1_{\mathcal{F}} \varepsilon} \mathcal{F}$$

are the identity endomorphisms $1_{\mathcal{G}}$ & $1_{\mathcal{F}}$.

1) (*) on Verma's: by Thm in Sec 1 of Lec 7,

$$\mathcal{F}(\Delta_A(w \cdot \lambda_i)) \simeq \Delta_A(w \cdot \lambda_3), \quad \mathcal{G}(\Delta_A(w \cdot \lambda_3)) \simeq \Delta_A(w \cdot \lambda_i).$$

By the universal property of the Verma modules $\text{End}_{\mathcal{O}_A}(\Delta_A(\mu)) \xrightarrow{\sim} \Delta(\mu)_\mu = A \quad \forall \mu \in \Lambda$. So $\varphi_{\Delta_A(w \cdot \lambda_3)}$ & $\varepsilon_{\Delta_A(w \cdot \lambda_i)} \in A$. By (**), their product is 1 finishing the proof in this case.

2) (*) on proj've generators. Thx to ($\heartsuit$), $\mathcal{O}_{w \cdot \lambda_i}$ admits a projective generator, $P_i := H_i \Delta_A(\lambda_3)$. It's filtered by Vermas b/c $\odot_i$ & $\mathcal{T}_{\lambda' \rightarrow \lambda_i}$ are exact functors sending Vermas to objects filtered by Vermas. The claim that ε_{P_i} is an isomorphism now follows from Case 1). Similarly, φ_{P_3} is an isomorphism.

3) (*) on arbitrary objects. Say, ε_{M_i} is an isomorphism thanks to a presentation $P_i^{\oplus m} \rightarrow P_i^{\oplus n} \rightarrow M_i \rightarrow 0$, case 2) ($\Rightarrow \varepsilon_{P_i^{\oplus ?}}$ is iso) & 5-lemma. This yields (*).

$\mathcal{T}_{\lambda_3 \rightarrow \lambda_2} \circ \mathcal{T}_{\lambda_1 \rightarrow \lambda_3}$ is an equivalence $w \Delta_A(w \cdot \lambda_1) \mapsto \Delta_A(w \cdot \lambda_2) \quad \square$

2) Structure of infinitesimal blocks.

Our goal now is to investigate the structure of the abelian categories $\mathcal{O}_A^\Xi$ & their projectives. We keep the assumptions of Sec 1.

2.1) Equivalence w. module category

We know, Sec 2.2 of Lec 5 that

$$\mathrm{Hom}_{\mathcal{O}_A}(M, M') \text{ is fin. gen}/A \quad \forall M, M' \in \mathcal{O}_A^\Xi \quad (1)$$

& that

$$\mathcal{O}_A^\Xi \text{ contains a projective generator, } P \quad (2)$$

Set $E := \mathrm{End}_{\mathcal{O}_A^{\mathrm{pr}}}(P)^{\mathrm{opp}}$. The following is standard over fields.

Proposition: We have the functor

$$\mathcal{F} := \mathrm{Hom}_{\mathcal{O}_A^{\mathrm{pr}}}(P, \cdot) : \mathcal{O}_A^{\mathrm{pr}} \longrightarrow E\text{-mod}_{\mathfrak{fg}}^{\text{fin. gen'd over } E}$$

& it's an equivalence

Sketch of proof:

By (1), $\forall M \in \mathcal{O}_A^\Xi \Rightarrow \mathrm{Hom}_{\mathcal{O}_A}(P, M)$ is fin. gen./A, hence $/E$

4 $\leadsto \mathcal{F}$. Also, E is finitely generated over A so is Noethe-

man. Hence every $N \in E\text{-mod}_{fg}$ is the cokernel of some $E^{\oplus m} \rightarrow E^{\oplus n}$. So we have the functor $G := P \otimes_E ? : E\text{-mod}_{fg} \rightarrow \mathcal{O}_A^{\vec{\Sigma}}$, it's left adjoint to F . We need to show that $\eta: id \rightarrow FG$ & $\varepsilon: GF \rightarrow id$ are isomorphisms. Note that η_E & ε_P are isomorphisms by constructions.

Note that F is right exact b/c P is projective. Now we use that $\forall N \in E\text{-mod}_{fg}$ (resp. $\forall M \in \mathcal{O}_A^{\vec{\Sigma}}$) is the cokernel of some $E^{\oplus m} \rightarrow E^{\oplus n}$ (resp. $P^{\oplus m} \rightarrow P^{\oplus n}$) & apply the 5-lemma to complete the proof. $\square$

Example: Let $\vec{\Sigma} = \{-\rho\}$. Then $\Delta_A(-\rho)$ is projective (as $-\rho$ is X -dominant) & it's a generator b/c every object in $\mathcal{O}_A^{\vec{\Sigma}}$ is filtered by quotients of $\Delta_A(-\rho)$. We have $E = A$ (cf. 1) in the proof of Thm in Sec 1), so

$$A\text{-mod}_{fg} \xrightarrow{\sim} \mathcal{O}_A^{\{-\rho\}}, N \mapsto \Delta_A(-\rho) \otimes_A N.$$

2.2) Freeness

5 Let $\vec{\Sigma} \in \Lambda/(W, \cdot)$ & $P = \mathcal{H}\Delta_A(\lambda')$ be as in ($\heartsuit$) in Sec 1.

Lemma: The algebra $\text{End}_{\mathcal{O}_A^\Xi}(P)$ is free as an A -module.

Proof: Since $\mathcal{Q}_i, T_{\lambda' \rightarrow \lambda}$ and their biadjoints $(\mathcal{Q}_i, & T_{\lambda \rightarrow \lambda'})$ are exact & send Vermas to objects filtered by Vermas we get a biadjoint H^* to H & $H^*H\Delta_A(\lambda')$ is filtered by Vermas (in $\mathcal{O}_A^{w \cdot \lambda'}$). Note that

$$\text{End}_{\mathcal{O}_A^{w \cdot \lambda}}(H\Delta_A(\lambda')) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_A^{w \cdot \lambda}}(\Delta_A(\lambda'), H^*H\Delta_A(\lambda'))$$

As was shown in Sec 2.1 of Lec 5, since λ' is dominant, $\text{Hom}(\Delta_A(\lambda'), M) \xrightarrow{\sim} M_{\lambda'}, \forall M \in \mathcal{O}_A^{w \cdot \lambda'}$.

So it suffices to show that all weight spaces in $H^*H\Delta_A(\lambda')$ are free over A . This is because the functor $M \mapsto M_\lambda: \mathcal{O}_A \rightarrow A\text{-Mod}$ is exact & weight spaces of Vermas are free & $H^*H\Delta_A(\lambda')$ is filtered by Vermas $\square$

2.3) Indecomposable projectives

We assume, in addition, that A is complete:

$$A \xrightarrow{\sim} \varprojlim_{k \rightarrow \infty} A/\mathfrak{m}^k$$

6] For $K = A/\mathfrak{m}$, we can view $\mathcal{O}_K^\Xi$ as a full subcategory of

all objects in $\mathcal{O}_A^\Sigma$ annihilated by m . Our goal is to establish:

Proposition: There are projectives $P_A(\lambda)$, $\lambda \in \Sigma$ s.t.

$$1) \operatorname{Hom}_{\mathcal{O}_A^\Sigma}(P_A(\lambda), L_K(\mu)) \simeq K^{\oplus \delta_{\lambda, \mu}} \quad \forall \lambda, \mu \in \Sigma$$

$$2) \quad \forall \text{ projective } P' \in \mathcal{O}_A^\Sigma, \text{ we have } P' \simeq \bigoplus_{\lambda \in \Sigma} P_A(\lambda)^{\oplus m_\lambda} \text{ where}$$

$$m_\lambda := \dim_K \operatorname{Hom}_{\mathcal{O}_A^\Sigma}(P', L_K(\lambda)).$$

Note that $\operatorname{Hom}_{\mathcal{O}_A^\Sigma}(P', L_K(\lambda))$ is an A -module killed by m (b/c $L_K(\lambda)$ is) so a K -vector space.

In the proof we'll need the following general result (lifting of idempotents)

Lemma: Let B be an associative unital ring, $I \subset B$ a nilpotent 2-sided ideal ($\exists n > 0 \mid I^n = 0$) & $\underline{\varepsilon} \in B/I$ an idempotent. Then $\exists$ idempotent $\varepsilon \in I$ with $\underline{\varepsilon} = \varepsilon + I$.

Proof: **exercise**, hint: reduce to $I^2 = 0$.

Proof of Proposition: take a projective generator P in $\mathcal{O}_A^\Xi$ & set $E = \text{End}(P)^{\text{opp}}$. Then $\mathcal{O}_A^\Xi \xrightarrow{\sim} E\text{-mod}_{fg}$, as in Sec 2.1.

Step 1: For any A -module N , & any fin. gen. projective P_E over E , we have $\text{Hom}_E(P_E, N \otimes_A ?) \xrightarrow{\sim} N \otimes_A \text{Hom}_E(P_E, ?)$.

For $k > 0$, set $A_k = A/\mathfrak{m}^k$, $E_k = A_k \otimes_A E$, etc. So

$$\text{Hom}_{E_k}(P_{E,k}, M_k) \xrightarrow{\sim} A_k \otimes_A \text{Hom}_E(P_E, M) \quad \forall M \in E\text{-mod}_{fg}$$

Note that $\mathcal{O}_k^\Xi \xrightarrow{\sim} E_k\text{-mod}_{fg}$.

Step 2: $\dim_K E_1 < \infty$, so $E_1 / \text{Rad}(E_1) \simeq \bigoplus_{\ell=1}^r \text{Mat}_{n_\ell}(\mathbb{D}_\ell)$, where $\mathbb{D}_1, \dots, \mathbb{D}_\ell$ are the endomorphism algebras of E_1 -irreps, equivalently $L_\kappa(\lambda)$'s, $\lambda \in \Xi$. We have $\text{End}_{\mathcal{O}_\kappa}(L_\kappa(\lambda)) \xrightarrow{\sim} K \quad \forall \lambda$ (exercise) $\Rightarrow \mathbb{D}_i = K \quad \forall i$. Let $\underline{\varepsilon}^i$ be primitive idempotents in

$\text{Mat}_{n_i}(K)$, $i = 1, \dots, \ell$. Applying Lemma to $B := E_1$, $I = \text{rad}(E_1)$,

we get idempotents $\varepsilon_1^1, \dots, \varepsilon_1^\ell \in E_1$. Then we construct

$\varepsilon_k^1, \dots, \varepsilon_k^\ell \in E_k$ inductively by lifting ε_{k-1}^i to ε_k^i again using the lemma. Since $E = \varprojlim E_k$ we get idempotents $\varepsilon^i \in E$.

Set $P^i = E \varepsilon^i$ so that

$$\text{Hom}_E(P^i, K^{\eta_j}) = \varepsilon^i K^{\eta_j} = \underline{\varepsilon}^i K^{\eta_j} \simeq K^{\oplus \delta_{ij}}, \text{ where we}$$

write K^{n_j} for the irrep of the summand $\text{Mat}_{n_j}(K)$.

If $L_K(\lambda)$ corresponds to K^{n_j} , write $P_A(\lambda) \in \mathcal{O}_A^{\Sigma_j}$ for the image of P^j under $\mathcal{O}_A^{\Sigma} \xrightarrow{\sim} E\text{-mod}_{fg}$.

Step 3: Proposition over K is classical, so we have an epimorphism $P' \twoheadrightarrow P'_1 \xrightarrow{\sim} \bigoplus_{\lambda \in \Sigma} P_A(\lambda)^{\oplus m_\lambda}$. Since P' is projective this lifts to $P' \xrightarrow{\varphi} \bigoplus_{\lambda \in \Sigma} P_A(\lambda)^{\oplus m_\lambda}$ that is an epimorphism by Nakayama lemma (to apply it we use the equivalence $\mathcal{O}_A^{\Sigma} \rightarrow E\text{-mod}_{fg}$ so that all modules involved are fin.gen./ A). Similarly we get an epimorphism

$$\bigoplus_{\lambda \in \Sigma} P_A(\lambda)^{\oplus m_\lambda} \xrightarrow{\psi} P'$$

Since $P', \bigoplus_{\lambda} P_A(\lambda)^{\oplus m_\lambda}$ are Noetherian, both φ & ψ are isomorphisms. $\square$

Exercise: P' is a generator $\Leftrightarrow m_\lambda > 0 \ \forall \ \lambda \in \Sigma$.